DETECTING OVARIAN ENDOMETRIOMAS FROM ULTRASOUND USING VISION TRANSFORMERS AND CROSS-MODALITY TRANSFER LEARNING

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ABSTRACT

Endometriosis is a chronic condition that affects around 10% of reproductive age women worldwide. It requires life-long management and, due to symptom overlap with other gynaecological conditions and the traditional requirement for invasive laparoscopy for diagnosis, there are often severe delays between becoming symptomatic and diagnosis. During this time patients often experience debilitating symptoms that can have a severe effect on their quality of life. Recently, standardised ultrasound approaches have been developed to aid diagnosis without the need for laparoscopy. However, these rely on trained specialists to acquire and analyse the ultrasound images - leading to bottlenecks in clinical pathways. In this paper, we present the first machine learning (ML) model trained on publicly available ultrasound data for endometrioma detection, demonstrating high performance (average precision: 0.81) despite heavily imbalanced data. Acknowledging that the limited amount of publicly available ultrasound data is hampering progress in this area, we investigate transfer learning using other (non-ultrasound) radiographic imaging modalities and find that this substantially improves endometriosis detection performance, despite the remaining significant data shift to ultrasound. This extends existing transfer learning literature in medical imaging, evidencing that medical imaging (non-ultrasound) models must learn low-level feature/pattern recognition that are applicable to unseen imaging modalities and body parts. Using SHAP values, we perform exploratory analyses of the features learned by our models and find no evidence of shortcut learning or bias. This study pioneers the use of open-source data and models in this domain, and we hope it motivates future collaborative, open-source efforts to refine and evaluate machine learning approaches for endometriosis.

1 Introduction

Endometriosis is a chronic inflammatory condition that is estimated to affect 10% of reproductive age women worldwide [1]. The condition requires life-long management and is associated with several debilitating symptoms that can have a severe effect on the patient's quality of life, including infertility, pain, and fatigue, and is also associated with numerous co-morbidities [2]. There is often a long diagnostic delay after the appearance of the first symptoms, due to significant symptom overlap with other gynaecological conditions, and symptoms often persist despite treatment. Not only does this delay in diagnosis severely affect patient's quality of life, it also places a substantial burden on the healthcare system due to repeat attendance to hospital until a proper diagnosis and effective treatment plan is put into place [3, 4]. The condition is estimated to cost the UK £12.5billion a year in healthcare and loss of work costs [5].

Definitive, gold standard diagnosis of endometriosis is via invasive laparoscopic surgery that visualises the lesions present, sometimes combined with histopathological confirmation [2]. However, due to its invasive, costly, and risky

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nature, access to this type of diagnosis is limited to those with sufficiently severe symptoms, meaning diagnosis is often delayed. Additionally, there are further biases in the access of this type of healthcare: socioeconomic and geographic barriers exist to access to laparoscopic surgery, and this has been shown to be further compounded by stigma, racism and/or elitism [6, 7].

A non-invasive diagnostic technique for endometriosis would democratise its diagnosis, however one is yet to exist. While ultrasound and/or MRI can be used to clinically diagnose endometriosis in some cases, absence of diagnostic markers in these imaging modalities does not rule out diagnosis as superficial lesions can be missed [8]. Specialist transvaginal ultrasound (TVUS) using the IDEA framework and its derivatives [9] have been shown to improve the performance of these non-invasive diagnostic techniques [10], but these still require specialist training to perform.

In other diagnostic imaging modalities, machine learning (ML) and artificial intelligence (AI) models have been shown to match, or even outperform, human expert diagnostic performance. For example, models such as convolutional neural networks (CNN) and vision transformers (ViT) have been shown to outperform primary care practitioners in skin cancer diagnosis from dermatoscopic images [11], detect diabetic retinopathy from fundus photographs [12], and match expert level accuracy in diagnostic morphology [13].

Similarly, AI has been applied to the task of endometriosis diagnosis. Ensemble approaches have been developed that use transvaginal ultrasound images for endometrioma segmentation [14] (Dice coefficient 0.767), and a ConvNext model has been trained to discriminate between benign and endometriosis cysts (AUROC: 0.90). However, these analyses are limited, using small datasets (limiting their statistical power [15]) and have little-to-no external validation, bias analysis, or generalisation performance evaluation which would be crucial for their deployment in clinical settings [16]. Furthermore, they all use private datasets of ultrasound images, limiting the reproducibility of their work. This makes them unsuitable for clinical use.

Indeed, due to the protracted length of endometriosis diagnostic pathways (in many cases, patients will have had many years of symptoms prior to final diagnosis), public endometriosis datasets are limited and small in sample size. While there are larger-scale endometriosis datasets such as GLENDA [17] and PelvicMRI [18], these use images/videos from laparoscopic surgery and fMRI data, respectively. Models trained on these datasets would not support more efficient diagnostic pathways, as patients would still have to undergo invasive and time-consuming surgery or costly MRI scans.

Conversely, ML models that can detect endometriosis from ultrasound images would meaningfully impact clinical pathways, as ultrasounds are cheap to perform but currently require specialists to analyse for endometriosis diagnostic. However, publicly available data in this area is limited to the MMOTU-2D [19] dataset which provides 1,469 ultrasound scans with annotated and labelled tumours collected from 294 patients.

Similarly, there are very few large-scale, public ultrasound datasets available for use by ML models, and even fewer still that are focused on gynaecological problems. Therefore, we investigate how transfer learning from *other*, non-ultrasound medical imaging modalities affects model performance on endometriosis detection. Although the benefits of transfer learning, especially in medical imaging, have been widely researched [20] they focus on models performing similar tasks on similar modalities (e.g., transfer learning across x-ray acquisition setups and tasks [21]). This is not practical with TVUS images, as publicly available ultrasound datasets are relatively small-scale and often do not cover gynaecological regions/organs, and ultrasound images exhibit significant data shift when compared to other medical imaging modalities such as x-rays or MRIs.

In this paper, we develop ML models which can accurately detect endometriosis in the MMOTU-2D dataset. We evaluate the effect pre-training and transfer learning has on vision transformer image classification models in the task of endometriosis cyst detection from these TVUS images, highlighting that even pre-training on different medical imaging modalities (e.g., x-rays) can improve performance in endometriosis detection over models trained on general image corpora. This is significant as it suggests this medical pre-training must encourage models to learn low-level feature/pattern recognition that is generalisable across unseen (and significantly different) imaging modalities and organs, which could be used to support ML model development in other low-resource domains.

We analyse the clinical net-benefit our models could have on the stratification of endometriosis diagnostic pathways [22], showing that they significantly improve upon the default treatment pathways. We then utilise post-hoc explainability techniques to elucidate areas of the ultrasound images the AI models placed most importance on. Finally, we discuss what this means for the future of AI-assisted endometriosis detection and propose a concrete set of next-steps required for future work in this area.

2 Methods

We utilise the MMOTU-2D dataset [19] for a binary endometriosis/non-endometriosis classification task. In this dataset, tumours are annotated with one of the following 7 labels: chocolate cyst, serous cystadenoma, teratoma, theca cell tumour, simple cyst, mucinous cystadenoma, high grade serous. Ultrasound images with no tumour present are labelled as 'normal ovary'. Although this dataset was collected with the wider task of tumour segmentation in mind (i.e., not limited to solely endometriosis), it includes a small number of patients (336 images) with diagnosed endometriosis cysts. In an endometriosis diagnostic pathway, tumour segmentation from ultrasound images is not necessary; a binary endometriosis/non-endometriosis flag alone is sufficient to progress the patient along the correct clinical pathway (either fast-tracked definitive diagnosis via laparoscopy or continuing treatment/diagnostics for other gynaecological issues).

Hence, we convert the tumour segmentation task from the MMOTU-2D dataset into a binary image classification task using the following labelling method, as chocolate cysts (endometrioma) are most often seen in the late stages of endometriosis [2]:

$$y(x) = \begin{cases} \text{endometriosis}, & \text{label} == \text{chocolate cyst} \\ \text{non-endometriosis}, & \text{otherwise} \end{cases}$$

As a single patient in the MMOTU-2D has multiple images, we use the train/test split released by the dataset authors, resulting in 1000 images in the training set and 469 in the test set, ensuring images from a single patient are contained within a single split. Due to the small size of the training set, we apply data augmentation techniques (random crops and horizontal flips) to increase the size of the training set and improve the generalisability of the trained model, and all images were resized to 224x224 and normalised as required by the ML model(s).

Following previous work showing their high performance and generalisability across medical image classification tasks [23], we use a 224x224 Vision Transformer (ViT) [24] as the base model for all experiments. To evaluate the effect domain-specific pre-training has on downstream classification performance, we compare a ViT model pre-trained on the general-corpus ImageNet [25] with a ViT (OmniRad-Small) that has been further pre-trained on 1.2 million radiology images [26]. OmniRad-Small was chosen over other medically pre-trained image foundation models as it could be confirmed that its pre-training did not include the MMOTU-2D dataset, meaning no data leakage could occur during model training.

Due to the heavily imbalanced nature of our labelling we also compare normal cross entropy loss with weighted cross entropy loss, with weights calculated using the formula $w_i = \frac{N}{2 \cdot n_i}$ where w_i is the weight for class i, N is the total number of samples, and n_i is the total number of samples in class i. Following guidance for the fine-tuning of ViTs [24, 26], across all experiments we use a learning rate of 5×10^{-5} for 10 epochs (with early stopping if necessary) and a batch size of 32 with gradient accumulation every 4 steps (e.g, an effective batch size of 128).

For evaluation on the test set, we focus on average precision (AP, i.e., the area under the precision-recall curve) as this is more robust to class imbalance. However, for completeness we also include the F1 score, precision, recall, and accuracy under the receiver operating characteristic curve (AUROC). To evaluate the clinical net-benefit of the model's, we apply decision curve analysis [22] to our models (on the test set). Decision curve analysis calculates the 'net benefit' of the model, defined as $M_{\text{sens}} \times \rho - (1 - M_{\text{spec}}) \times (1 - \rho) \times \omega$, where M_{sens} is sensitivity, M_{spec} is specificity, ρ is the disease prevalence, and ω is the odds at the given threshold probability. As decision curve analysis incorporates the consequences of the decision made as a result of the model's predictions, it better captures the clinical utility of a model/test when compared to traditional ML metrics such as AP and AUROC. To explore post-hoc explainability of our best performing model, we utilise GradientSHAP [27] using the all-zero tensor as the baseline, choosing GradientSHAP for its efficient estimation of SHAP values.

3 Results

After applying our labelling technique to the MMOTU-2D dataset, 336 images (22.9%) have the positive endometriosis label: 226 in the training set and 110 in the test set. Table 1 reports the fine-tuned model performance on the MMOTU-2D test set.

Figure 1 breaks down model performance by model and class (endometriosis/non-endometriosis) and Figure 2 shows ROC and PR curves for each of these models. Net benefit curves for all 4 models are compared with the default treatment options of treating all patients or treating no patients, in Figure 3. All four models demonstrated greater net benefit than the treat-all strategy above a decision threshold of 0.05, and better than the treat-none strategy below a high

Table 1: Performance of all models on the MMOTU-2D test set. **Bold** indicates the best performing for that metric.

Model	AUROC	AP	Precision	Recall	F1
Vision Transformer	0.836	0.677	**0.826**	0.509	0.629
Vision Transformer (Weighted Loss)	0.911	**0.813**	0.790	0.718	0.752
OmniRad-Small (Weighted Loss)	**0.923**	0.808	0.761	**0.782**	**0.771**
OmniRad-Small	0.908	0.807	0.814	0.718	0.763

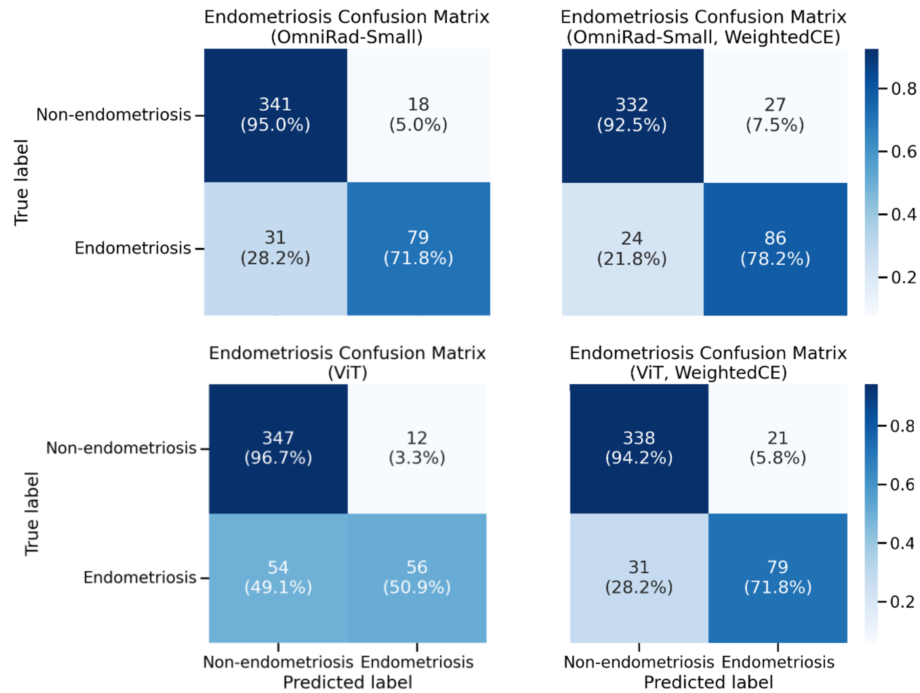

Figure 1: Confusion matrices on the MMOTU-2D test set for all models evaluated.

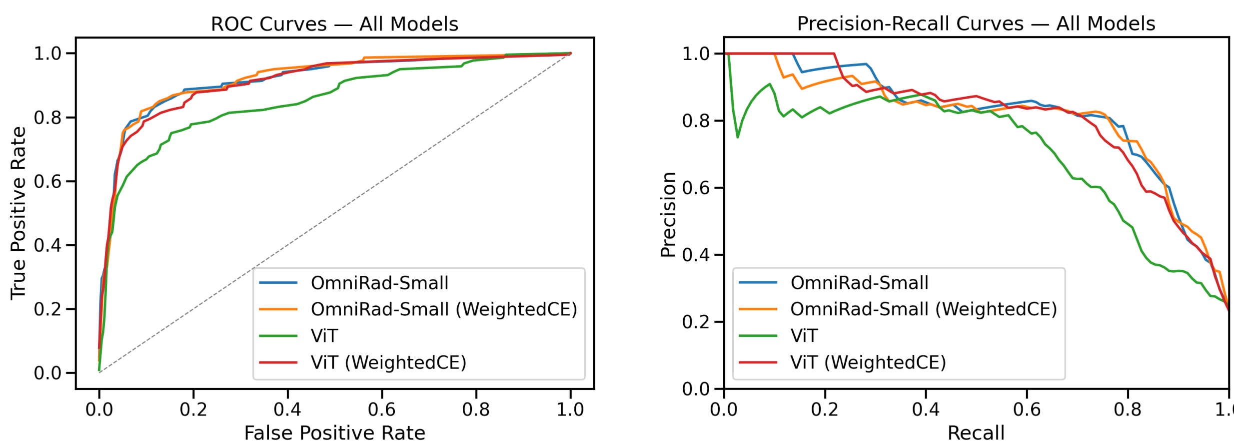

Figure 2: Combined plots showing (a) Receiver Operating Characteristic, and (b) Precision-Recall curves for all models evaluated.

diagnostic threshold of 0.83. As in the traditional analyses of model performance in Figure 2, Figure 3 highlights that the OmniRad-Small models outperformed the non-medically finetuned ViT-Small variants.

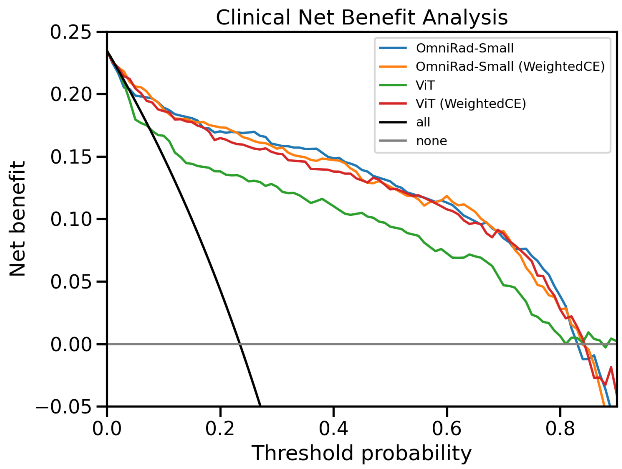

Figure 3: Net benefit curves from the decision curve analysis of all ML models, compared with the default strategies of treating everyone (all) or none.

As OmniRad-Small trained with weighted cross entropy loss achieves the highest levels of performance (AUROC, recall, F1, clinical net benefit), we run GradientSHAP on it across 8 random samples from the test set (4 with non-endometriosis ground truth labels, and 4 with endometriosis labels), the results of which are shown in Figure 4.

4 Discussion

Across both the regular, ImageNet pre-trained Vision Transformer and OmniRad, fine-tuning with a weighted cross entropy loss substantially improved performance, highlighting the need to handle class imbalances when present in the data. This is demonstrated by the higher precision of the regular ViT model using non-weighted cross entropy loss, which is let down by the much lower recall, indicating the model has a high false negative rate (also see Figure 1, bottom left, and Figure 2).

Additionally, we found that OmniRad outperforms a regular ViT when fine-tuned on the MMOTU-2D dataset. Although this follows from previous work on transfer learning in the medical imaging domain [28], it is interesting to see a similar trend here despite OmniRad's pre-training corpus consisting of mainly other radiography image modalities (i.e., it is mainly x-ray, CT, and MRI images). This highlights the significant domain shift between the ImageNet pre-training of the baseline ViT and the MMOTU-2D dataset - moving to medical images in the OmniRad model softens the impact of this shift and suggests that the medical pre-training is learning low-level feature/pattern recognition that is generalisable across imaging modalities, and even on unseen anatomy. This opens up future work to explore more generalised medical imaging pre-training for other tasks, rather than the current approach of preferring intra-modality transfer learning, alleviating data availability issues. It would be interesting for future work to use advances in mechanistic interpretability to understand which pattern recognisers are retained during these transfer learning processes.

Decision curve analysis (Figure 3) highlights the significant impact automated endometriosis detection using these ML models could have on clinical practice, with all models out-performing the basic treat-all and treat-none strategies at clinically relevant thresholds. Even at higher thresholds (e.g., > 0.5), which may be relevant when the next step in the diagnostic pathway could be invasive laparoscopic surgery, our models outperform these baseline treatment strategies. Furthermore, at lower decision thresholds (e.g., $0 < t \leq 0.2$), which may be used if the next diagnostic step was more screening/triage, our best performing OmniRad-based models continue to out-perform these baseline options. This highlights the significant impact ML-based endometriosis detection could have on these clinical pathways.

For images of endometriosis, the SHAP plots in Figure 4 show that our best performing model (OmniRad with weighted cross entropy) places most of its importance around and in the endometrioma. However, there are some areas of note. It is encouraging that in the first, second and third endometriosis samples (Figure 4, first, second and third rows in the right column) there is little-to-no importance placed upon the annotations hard-coded into the ultrasound images by the healthcare practitioners capturing/analysing the scan (e.g., the green bounding box and red/green/blue highlights in the second sample, and the text added in the third sample). This suggests that the model has avoided learning spurious correlations and is instead focusing on clinically meaningful features. Whereas SHAP importance in the endometrioma samples is typically concentrated on the cysts themselves, a much wider spread of SHAP values is observed in samples with no endometrioma present (Figure 4, left hand column). This is to be expected, as the non-endometriosis class contains a wide variety of other conditions (including completely normal images) and thus the 'negative' features are also more varied. Similarly to the endometriosis samples, these also encouragingly show little importance placed around hard-coded annotations.

To the best of our knowledge, this is the first study to use publicly available data to train ML models for endometriosis detection from ultrasound images. While direct comparisons with previous work on ultrasound images for endometriosis diagnosis cannot be made, as they use private datasets, our results match and/or outperform previous studies in this area [29] (AUROC theirs: 0.90, ours: 0.92). Our thorough analysis of medical pre-training and weighted loss functions has clearly demonstrated the efficacy of domain-specific pre-training and highlights the potential of using other (i.e., non-TVUS) sources of ultrasound data for an even more domain-specific pre-training regimen.

Nevertheless, despite this potential, there are areas for future work. Bias and generalisability are key concerns in medical ML; for example, it has been shown that many ML models may be biased towards/against a certain population, and/or may not generalise to unseen patient populations (e.g., from a different hospital, or images acquired using different hardware) [30]. Due to the limitations of the MMOTU-2D dataset, which is collected from a single site and contains no patient demographics data, we were unable to analyse our model's bias or generalisability. Therefore, it would be prudent for future work to evaluate our techniques on external datasets prior to any prospective validation studies. Furthermore, while we have used SHAP explainability methods to inspect the model's learned features and confirm it is not using 'shortcuts', this is only a preliminary analysis. One may consider applying mechanistic interpretability (MI) techniques such as circuit analysis to better elucidate the inner workings of our models; however, this is a field in its infancy and requires medical imaging-specific MI techniques to be analysed, and so we leave this for future work. Finally, after the more extensive retrospective evaluation of our models on external datasets (mentioned above) has been completed, prospective studies need to be established to compare ML-assisted endometriosis diagnosis with current clinical pathways and to evaluate its cost-effectiveness and impact on patient care.

5 Conclusion

This study is, to the best of our knowledge, the first to use publicly available ultrasound data to train ML models for endometriosis detection. Through the development of high-performing models (AP: 0.81), we have shown that it is possible to train modern ML models for endometrioma detection, out-performing existing techniques developed on private datasets. Additionally, we have shown that using models pre-trained on other medical imaging modalities can substantially improve performance over baseline pre-trained image classifiers. Not only could this have a profound effect on endometriosis care worldwide, lowering the reliance on (resource-limited) specialist acquisition and analysis of medical images for endometriosis diagnosis, but it also opens the door to more general transfer learning corpora in other resource-limited settings.

For endometriosis detection, we believe our techniques would have the strong potential to not only significantly reduce the time-to-diagnosis for patients suffering from the condition but, by democratising access to diagnosis, also reduce the biases currently seen in endometriosis pathways. By lowering the time taken to diagnose, patients would hopefully receive treatment sooner, reducing their reliance on the health system and thus reduce their healthcare costs and improve their quality of life.

We hope that our use of open source datasets and models, and high-performing models for endometriosis detection using a comparatively small dataset, encourages future research to continue to extend our work as new and larger ultrasound datasets are released, as we believe model performance will only continue to improve.

Competing Interests

All authors declare no competing interests.

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Author Contributions

All authors contributed to the design and conceptualisation of the study. MW prepared and performed the analyses, results interpretations, and drafted the manuscript and figures. All authors critically reviewed and revised the manuscript.

Data Availabilty

The MMOTU-2D dataset [19] used in this study is available at `https://github.com/cv516Buaa/MMOTU_DS2Net#dataset`.

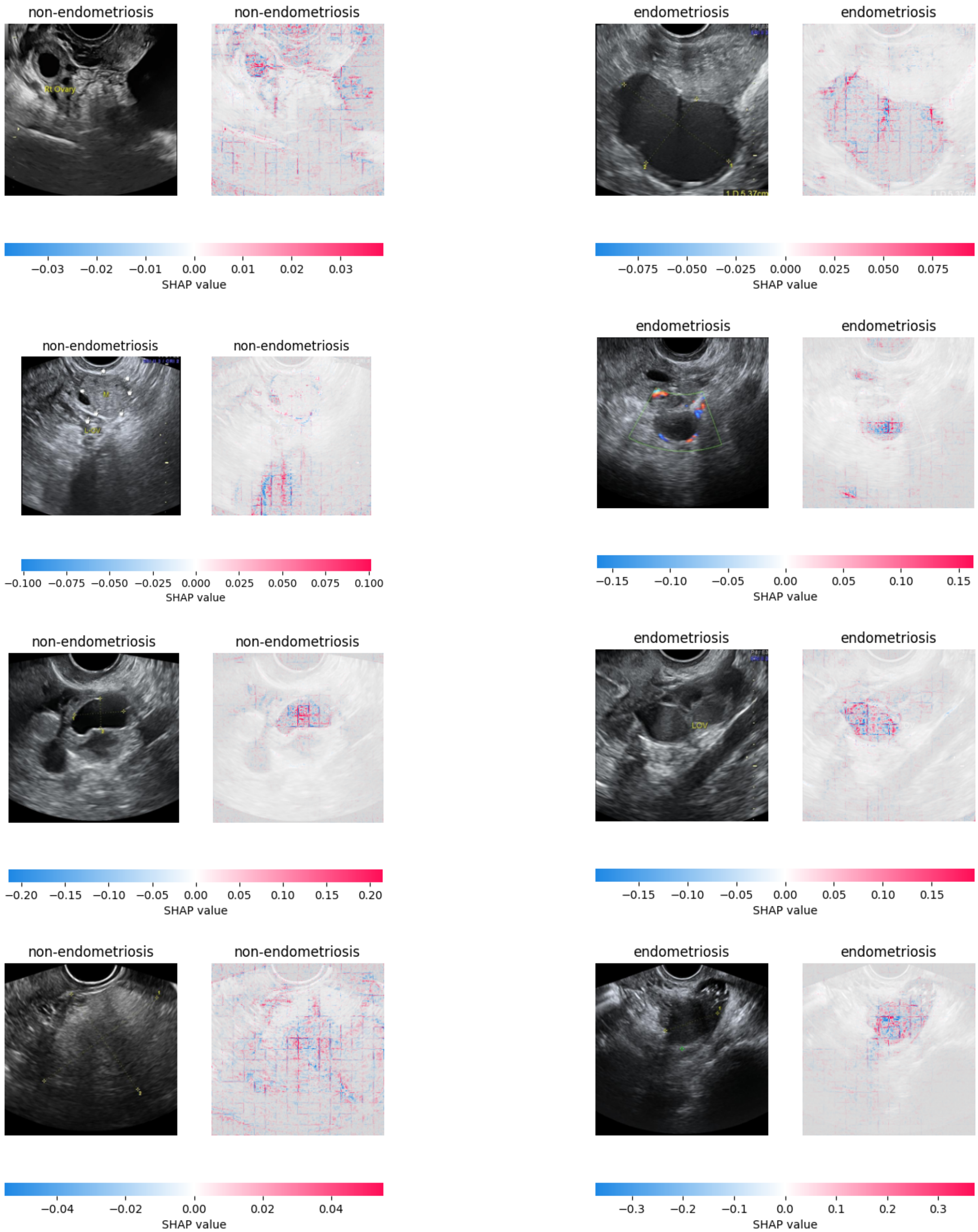

Figure 4: Comparison of SHAP values from the OmniRad-Small (WeightedCE) model on 8 random samples from the test set. For each pair of images, the left is the original and the right the SHAP values. The left column of image pairs are non-endometriosis ground truth samples, and the right are endometriosis samples.

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